## Oxides and Hydroxides

On combustion in excess of air, alkali metals forms normal oxides with formula $M_2O$. They react with water to form corresponding hydroxides which are basic in nature.

$$M_2O + H_2O \rightarrow 2\ MOH$$

Alkali metals apart from lithium form peroxides in addition to normal oxides upon combustion with excess air. These peroxides produce hydroxides and $H_2O_2$ upon reacting with water.

$$M_2O_2 + 2\ H_2O \rightarrow 2MOH + H_2O_2$$

(M = Na, K, Rb, Cs)

Except lithium and sodium, all the other alkali metals form superoxides also. These superoxides also gives basic hydroxides upon treatment with water.

$$2\ MO_2 + 2\ H_2O \rightarrow 2\ MOH + H_2O_2 + O_2$$

(M = K, Rb, Cs)

Under appropriate conditions pure compounds $M_2O$, $M_2O_2$ or $MO_2$ may be prepared.

## Properties of oxides and hydroxides:

The oxides and the peroxides are colourless when pure, but the superoxides are yellow or orange in colour. The peroxides are diamagnetic while the superoxides are paramagnetic. Sodium peroxide is widely used as an oxidising agent. The hydroxides which are obtained by the reaction of the oxides with water are all white crystalline solids. The alkali metal hydroxides are strong bases. They dissolve in water with evolution of heat on account of intense hydration.

## Halides:

The alkali metal halides, MX, (X=F, Cl, Br, I) are colourless crystalline solids with high melting points. They can be prepared by the reaction of the appropriate oxide, hydroxide or carbonate with aqueous hydrohalic acid (HX). As the electropositive character of alkali metal increases from Li to Cs, the ease with which the metals form halides increases from Li to Cs. All halides are ionic in nature except LiBr and LiI. Except LiF, all other halides are soluble in water. The low solubility of LiF in water is due to its high lattice enthalpy (small size of $Li^+$ and $F^-$). Due to the presence of covalent nature both LiBr and LiI are soluble in organic solvents.

## Salts of oxo-acids

Alkali metals form salts with all the oxo-acids. Most of these salts are soluble in water and are thermally stable. As the electropositive character increases down the group, the stability of the carbonates and bicarbonates increases. This is due to the decrease in polarising power of alkali metal cations. The carbonates ($M_2CO_3$) of alkali metals are remarkably stable up to 1273 K, above which they first melt and then eventually decompose to form oxides. However, $Li_2CO_3$ is considerably less stable and decomposes readily.

$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$$

This is presumably due to large size difference between $Li^+$ and $CO_3^{-2}$ which



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